Graph Traversal Algorithms: BFS and DFS

1. Breadth-First Search (BFS)

Definition: Breadth-First Search (BFS) is a graph traversal algorithm that visits vertices level by level. It visits all adjacent vertices of the current vertex before moving to the next level.

Data structure used: Queue (FIFO — First In, First Out)

Algorithm of BFS

```
BFS(Graph, Start)

1. Create an empty queue Q.
2. Mark Start as visited.
3. Insert Start into Q.

4. While Q is not empty, repeat:
     a. Remove a vertex V from Q.
     b. Visit and print V.
     c. For every adjacent vertex U of V:
          If U is not visited:
              Mark U as visited.
              Insert U into Q.

5. Stop.
```

Working: BFS first visits the starting vertex, then its immediate neighbours, followed by the neighbours at the next level.

2. Depth-First Search (DFS)

Definition: Depth-First Search (DFS) is a graph traversal algorithm that explores as far as possible along one path before backtracking to explore other paths.

Data structure used: Stack (LIFO — Last In, First Out) or recursion.

Algorithm of DFS (Recursive)

```
DFS(Graph, V)

1. Mark V as visited.
2. Visit and print V.

3. For every adjacent vertex U of V:
     If U is not visited:
         DFS(Graph, U).

4. Stop.
```

Working: DFS starts from a vertex, visits an unvisited neighbour, and continues deeper until no unvisited neighbour remains. It then backtracks.

3. Difference Between BFS and DFS

BFS	DFS
Breadth-First Search	Depth-First Search
Visits vertices level by level	Explores one path deeply first
Uses a queue	Uses a stack or recursion

BFS	DFS
Generally finds the shortest path in an unweighted graph	Does not necessarily find the shortest path
Useful for level-order exploration	Useful for exploring paths and connectivity

Important Exam Points

• BFS uses **enqueue and dequeue** operations.

• DFS uses **recursion** or **push and pop** operations when implemented with a stack.

• Both algorithms use a **visited array/set** to avoid visiting the same vertex repeatedly.

• For a disconnected graph, restart traversal from an unvisited vertex to visit every component.